

WEATHER DATA ANALYSIS & TEMPERATURE PREDICTION REPORT

Generated on 2026-06-15 18:24:14 | Dataset: seoul_weather_2022_to_2024.csv

1. DATASET OVERVIEW

This report presents a meteorological analysis of the dataset **seoul_weather_2022_to_2024.csv**. The dataset contains a total of **731** records and **33** variables.

Metric	Value / Column Mapping
Date Range	2022-01-01 to 2024-01-01
Detected Date Column	datetime
Target Variable (Temp)	temp
Humidity Column	humidity
Precipitation Column	precip
Wind Speed Column	windspeed
Conditions Column	conditions

2. WEATHER STATISTICS SUMMARY

Weather Metric	Average / Total Value	Maximum Recorded	Minimum Recorded
Temperature (°C)	13.69 °C	31.10 °C	-13.70 °C
Humidity (%)	65.28 %	99.00 %	22.00 %
Rainfall (mm)	4.35 mm (Avg)	268.62 mm	3178.11 mm (Total)
Wind Speed (m/s)	14.37 m/s	33.00 m/s	N/A

Weather Conditions: Most Common Condition is **Partially cloudy**, and the Least Common Condition is **Snow, Overcast**.

3. GRAPHICAL DATA ANALYSIS

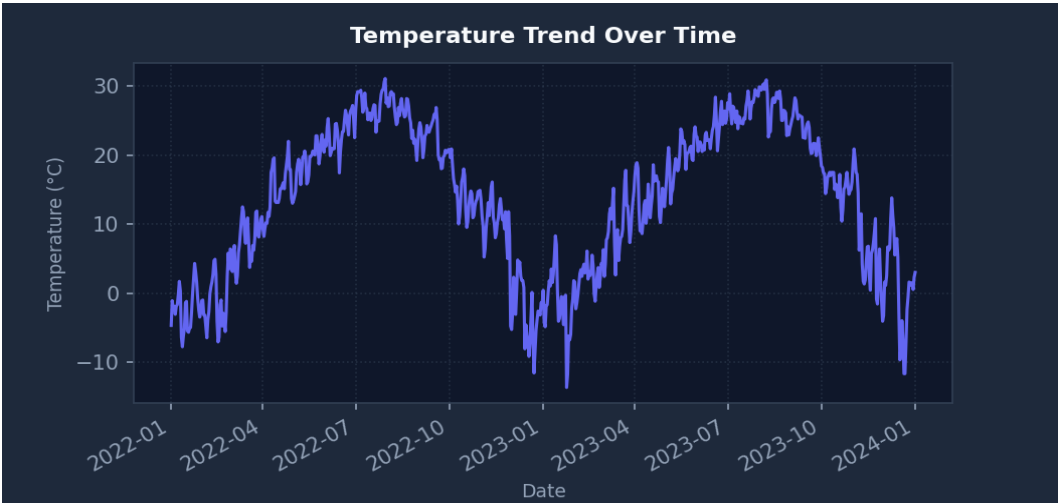


Figure 1: Chronological Temperature Trend

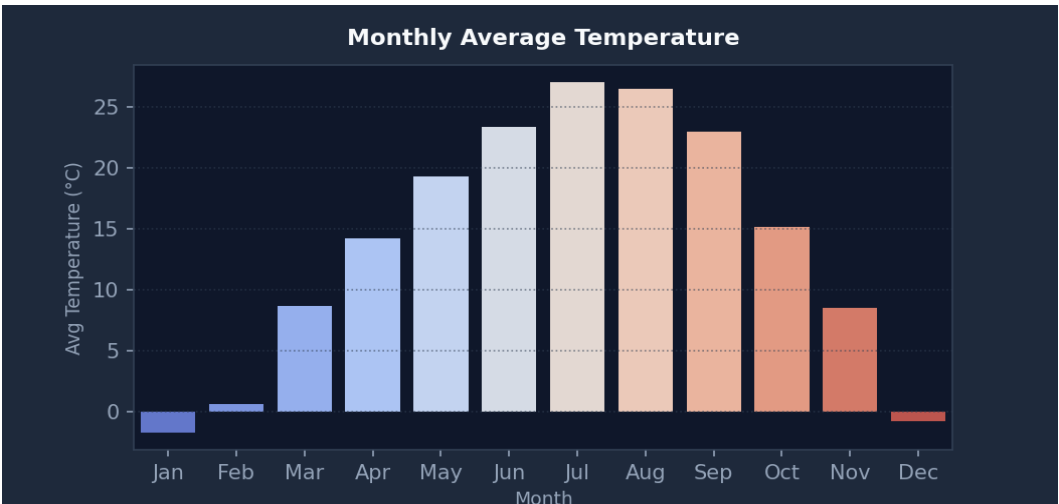


Figure 2: Monthly Average Temperature

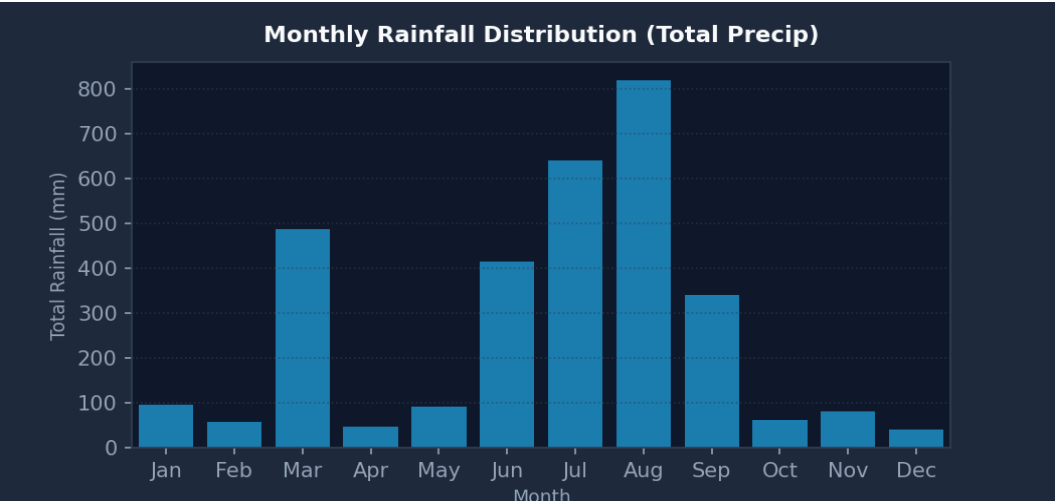


Figure 3: Monthly Rainfall Distribution

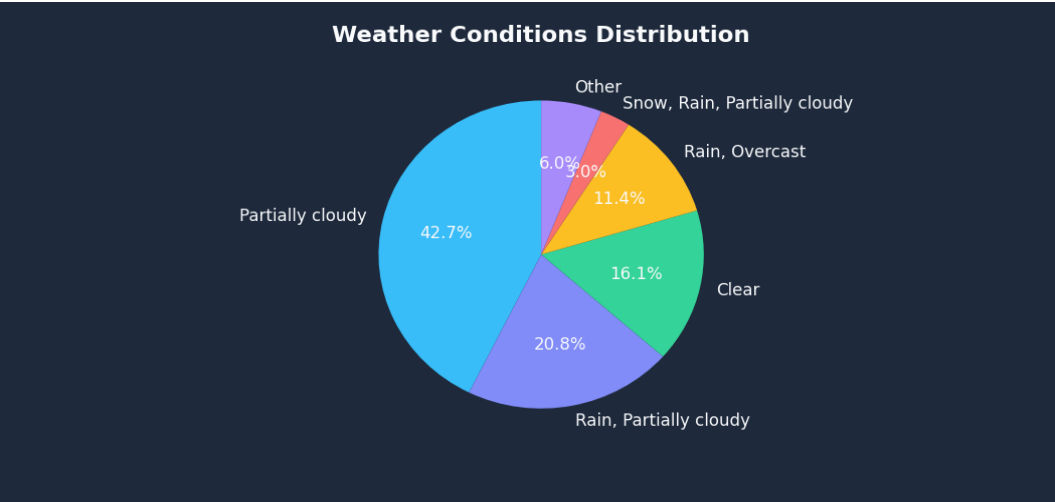


Figure 4: Weather Conditions Distribution

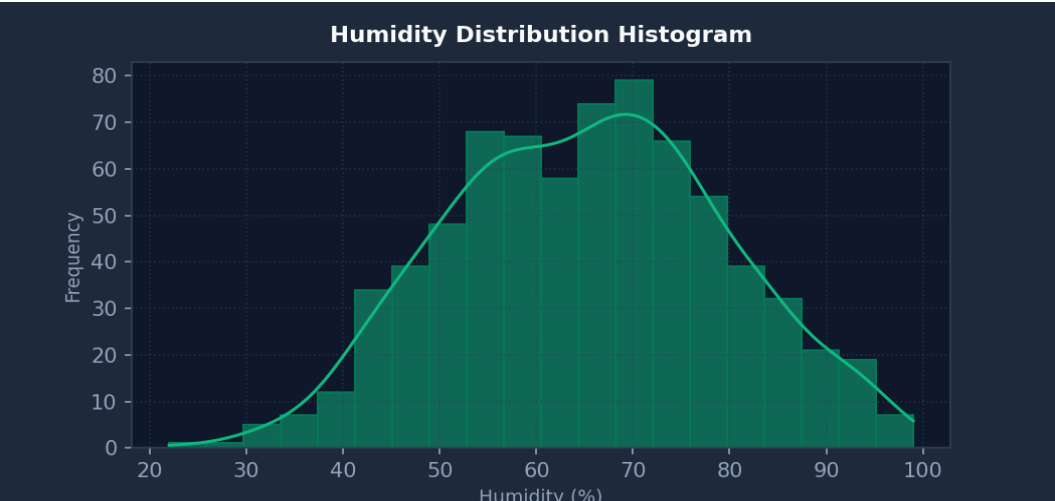


Figure 5: Humidity Distribution Histogram

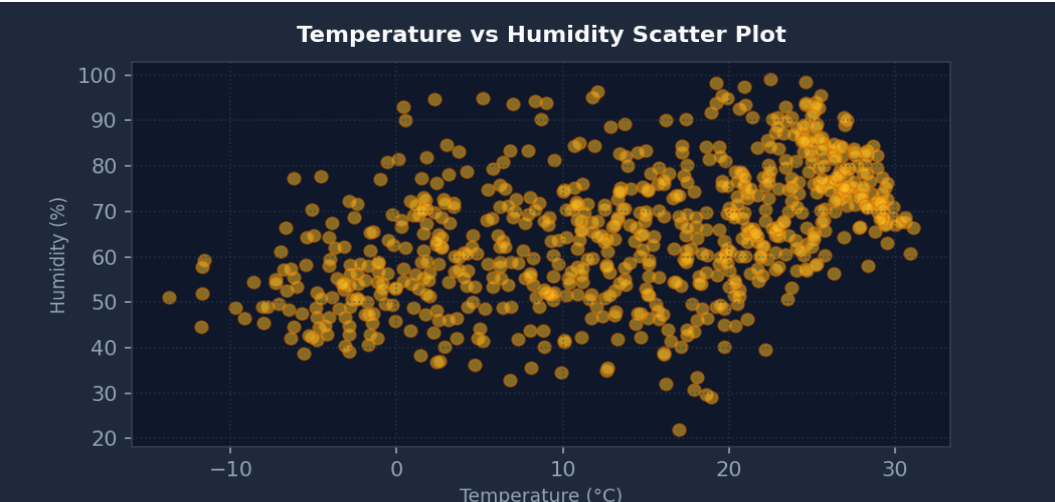


Figure 6: Temperature vs Humidity Correlation

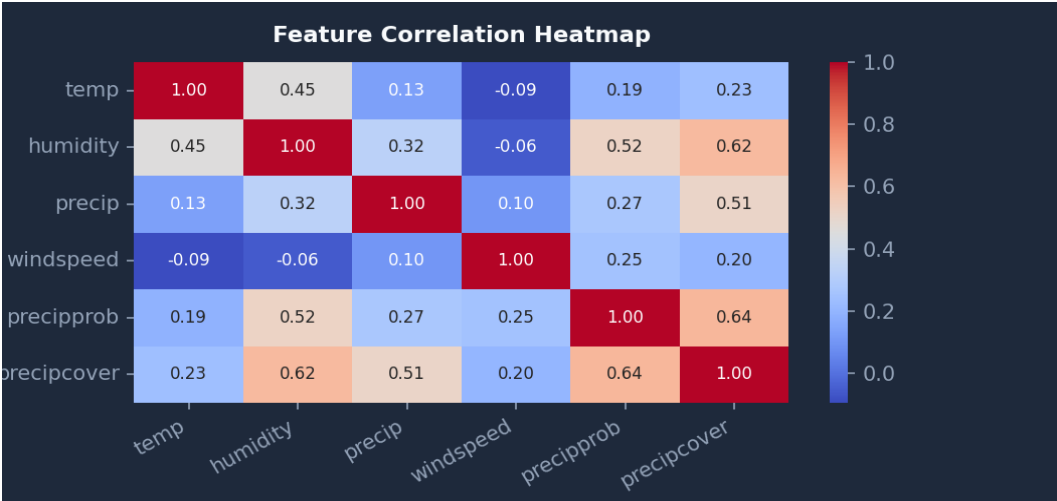


Figure 7: Feature Correlation Heatmap

4. MACHINE LEARNING RESULTS & PERFORMANCE

A **Linear Regression** model was trained to predict **temp** using the available numerical features. The dataset was split into **80% training set** (584 samples) and **20% testing set** (147 samples).

Evaluation Metric	Value	Interpretation
Mean Absolute Error (MAE)	4.1071 °C	Average magnitude of the errors in predictions.
Mean Squared Error (MSE)	27.5747	Average squared difference between predicted and actual values.
Root Mean Squared Error (RMSE)	5.2512 °C	Standard deviation of the residuals (prediction errors).
R² Score (R-squared)	0.7698	Proportion of target variance explained by the model features.

Model Coefficients Summary:

Predictor Feature	Coefficient	Effect on Temperature
precip	0.0249	Positive relationship (Temp rises)
precipprob	-0.0064	Negative relationship (Temp drops)
precipcover	0.0584	Positive relationship (Temp rises)
snow	-0.3882	Negative relationship (Temp drops)
snowdepth	-1.9296	Negative relationship (Temp drops)
windgust	-0.0850	Negative relationship (Temp drops)
windspeed	-0.2832	Negative relationship (Temp drops)
winddir	-0.0090	Negative relationship (Temp drops)
sealevelpressure	-0.7680	Negative relationship (Temp drops)
cloudcover	0.0496	Positive relationship (Temp rises)
visibility	0.1222	Positive relationship (Temp rises)
solarradiation	0.0473	Positive relationship (Temp rises)
solarenergy	-0.1196	Negative relationship (Temp drops)
severerisk	0.0640	Positive relationship (Temp rises)
moonphase	-1.0229	Negative relationship (Temp drops)
[Intercept Constant]	790.6059	Base temperature when all features are zero

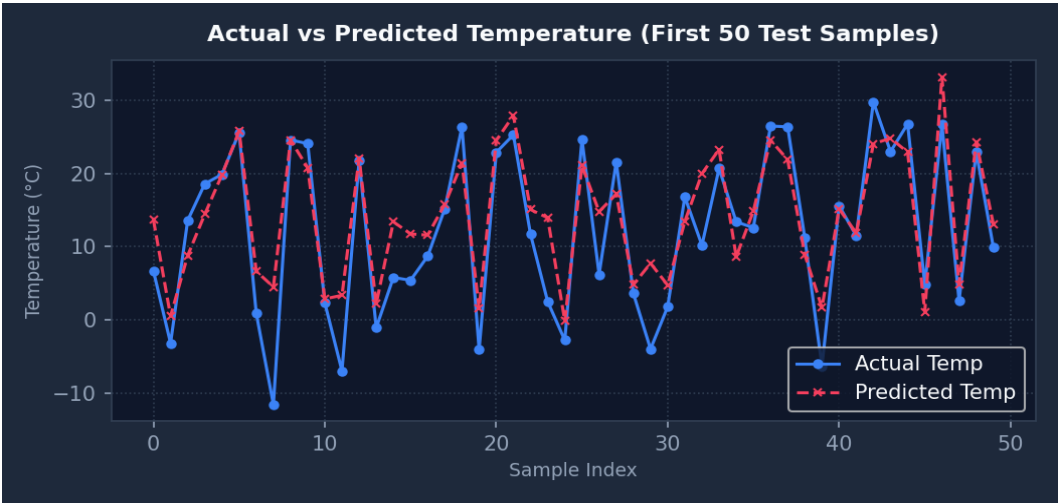


Figure 8: Actual vs Predicted Temperature on Test Set